Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/12/2011
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:	NOTE: Received funding continuation to 4/14/2013, per NSSC (8/2010) NOTE: Received extension to 2/28/11 (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AD22G		
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Task Description:

Fundamental physics research is carried out by performing experiments with dusty plasmas. Ground-based laboratory experiments are carried out to prepare for future microgravity flight experiments, including the planned European PK-4 instrument. Data were analyzed from the predecessor project PK-3 Plus on the International Space Station (ISS). (The acronym PK refers to "Plasma Crystal.")
Dusty plasma consists of particles of solid matter suspended in ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma.

Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We study:

• Structural features. These include the boundaries of the particle suspension, a void at the plasma's center, and crystalline or liquid-like microstructure in the arrangements of particles.

• Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, transport, and random motion associated with a non-equilibrium system. Dynamical behavior is studied at an atomistic level, using video micrography to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter.

Experimental methods include direct imaging of particle suspensions using video microscopy and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers.

Ground-based laboratory experiments and numerical simulations accompany the analysis of data returned from space. This ground-based work serves the dual purposes of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return. This ground-based work is mainly performed in the PI's laboratory at The University of Iowa, but it can also include analysis of data from parabolic-flight experiments conducted by Europeans.

The PK-4 mission is an ESA project, under the direction of ESA. The PI collaborates with the ESA scientist who is responsible for the mission. The PI is a member of ESA's Facility Science Team for the PK-4 mission.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Analysis of PK-3 Plus microgravity experiment was done by members of Professor Goree's group. We investigated non-Gaussian statistics, which is a fundamental statistical physics problem that can take great advantage of the unique data provided by dusty plasma experiments. Using data from a microgravity experiment with no macroscopic flows, we analyze the microscopic random motion of the charged dust particles, as they interact with one another. We calculate the velocity distribution function, and we find that it has "fat tails," i.e., an overabundance of energetic particles as compared to a Gaussian (Maxwell-Boltzmann) distribution. We fit the measured distribution to the Tsallis distribution, which is a theoretical expression intended for non-equilibrium systems, and we found a good fit. Such a result was found previously in laboratory experiments under conditions with high-speed ion flows; here, microgravity conditions enabled experiments with low-speed ions. (This is because in the presence of gravity, a large dc electric field is required to levitate the dust particles, and this large electric field drives a fast ion flow, while under microgravity conditions the electric field can be orders of magnitude weaker.) Because ion flow past the charged dust particles is a non-equilibrium condition, signatures of non-Gaussian statistics are expected, but these signatures are expected to be weaker with low-speed ions (in microgravity conditions) than with high-speed ions (in laboratory conditions). Surprisingly, we find statistically-significant deviations from Gaussian statistics, even with the low-speed ion flow under microgravity conditions.
During this period, Professor Goree's group also carried out ground-based laboratory research to support future flight experiments.

Dust density wave:

Two projects were studies of the dust density wave, which is a low-speed compression wave. It is expected to be observed in PK-4, so our laboratory experiments help prepare for planning PK-4 experiments by identifying experimental methods and data analysis methods. Two dust-density wave experiments were completed and reported:

• The first experiment was a study of nonlinearities in the dust-density wave. In this second project, we introduced several innovations to the study of nonlinear plasma waves. We focused our attention on harmonics of the fundamental wave as the signature of nonlinearity. Using our recently-developed phase-sensitive detection method, we found that the wavenumber and growth rate of each harmonic was surprisingly a simple integer multiple of those for the fundamental. We also introduced the total-harmonic distortion (THD) as a measure of wave nonlinearity, exploiting our measurements that were made possible by the phase-sensitive detection. Results from this second experiment were published in Physics of Plasmas.

• The second dust-density wave experiment was a characterization of nonlinear synchronization. In this phenomenon, a natural oscillation (like the dust density wave) that tends to occur at frequency f0 will shift its frequency if there is a periodic external disturbance at a different frequency fdrive. We carried out laboratory experiments and found several synchronized states, including the simple 1:1 state where the oscillation frequency shifts to match fdrive. We discovered features for this synchronization that have not been previously reported, including a threshold amplitude for the drive and a branching feature in the Arnold tongue diagram for the synchronization. This work was submitted to Physical Review E.

Instrumentation methods and data analysis:

• We performed and reported a project to improve an analysis method that will be used with PK-4 data to measure particle velocities using video imaging. This method, called Particle Tracking Velocimetry (PTV), uses particle positions measured in two consecutive video frames to calculate velocities. In this project we characterized the errors that occur, and how they vary with the camera frame rate. This is important because PK-4 experimenters will use PTV, and they will need to know how large their measurement errors are, and how to minimize these errors when they select the frame rate before the experiment. This project was reported in Review of Scientific Instruments.

Viscosity measurement:

• We performed and reported a project to develop a new way of measuring the viscosity of a liquid-like dusty plasma. Such a liquid is actually an arrangement of microspheres suspended in space in an irregular lattice that does not resist shear, and it is known to form in microgravity experiments. We formed the liquid in a laboratory experiment by first preparing a 2D crystal, and then melting it by applying laser manipulation. We measured viscosity in a new way, which has previously only been used for computer simulations and not with experimental data. This method is called the Green-Kubo method. Its inputs are positions and velocities of dust particles, which we measured using PTV. Using these measurements as inputs into the Green-Kubo relation for viscosity, we obtained a value for the viscosity. We verified that this value is consistent with values that have been previously reported using conventional methods. The advantage of this method is that it does not require an applied shear, as is the case for example with a rheometer. This is important because it is difficult to apply a shear to a dusty plasma since the dust particles are not in contact with any solid surface. We expect that this method can be used to determine viscosity in future PK-4 experiments.

Microgravity parabolic flight analysis:

• Microgravity data was analyzed for parabolic flights that were conducted by German collaborators. Results were reported in a paper submitted to Physics of Plasmas.

Numerical simulations were carried out to study viscoelasticity. These simulations were carried out in collaboration with two leading Hungarian scientists. Results were reported in a paper submitted to Physical Review E. The results of this effort are expected to be suitable for direct comparison to experiments.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Articles in Peer-reviewed Journals	Flanagan TM, Goree J. "Development of nonlinearity in a growing self-excited dust-density wave." Physics of Plasmas, 2011 Jan;18(1):013705. http://dx.doi.org/10.1063/1.3544938 , Jan-2011
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Articles in Peer-reviewed Journals	Feng Y, Goree J, Liu B. "Errors in particle tracking velocimetry with high-speed cameras." Review of Scientific Instruments, 2011 Mar;82(5): 053707. http://dx.doi.org/10.1063/1.3589267 , May-2011
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Articles in Peer-reviewed Journals	Feng Y, Goree J, Liu B, Cohen EGD. "Green-Kubo relation for viscosity tested using experimental data for a two-dimensional dusty plasma." Physical Review E, 2011 Oct;84(4):046412. http://dx.doi.org/10.1103/PhysRevE.84.046412 , Oct-2011
Papers from Meeting Proceedings	Donko Z, Goree J, Hartmann P. "Complex viscosity of 3D Yukawa liquids." International Conference on the Physics of Dusty Plasmas, Garmisch, Germany, May 16-20, 2011. DUSTY/COMPLEX PLASMAS: BASIC AND INTERDISCIPLINARY RESEARCH: Sixth International Conference on the Physics of Dusty Plasmas. AIP Proceedings vol. 1397, p. 307-307, 2011. http://dx.doi.org/10.1063/1.3659816 , Nov-2011